

Does Board Quality Influence Earnings Management Behaviour of Firms? Evidence from an Emerging Economy

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1. Introduction

Firms often resort to earnings management practices with a variety of motives.

Several factors may motivate firms to manage their earnings but 'market reaction' has remained as one of the most important factors (Friedlan, 1994; Erickson and Wang, 1999; Bartov et al., 2002). If reported earnings of a firm is markedly less than the estimated earnings, it may adversely impact its share prices and if this trend continues over the years, it may give rise to suspense in the market about the earnings stability of the firm. Hence, managers are often motivated to use earnings management to smooth out fluctuations in earnings performance (Barnea et al., 1976). Receiving increased amount of bonus, fear of government investigation and intervention may also motivate managers towards earnings manipulation (e.g. Watts and Zimmerman, 1978; Healy, 1985 Jones, 1991; Haw et al., 2005).

Firms under financial distress often manipulate earnings for several reasons like, contractual renegotiations and concealing of financial distress (e.g. DeAngelo et al., 1994; Burgstahler and Dichev, 1997; Rosner, 2003). Managers of a firm, which is under financial distress, may have their bonuses cut, loss of reputation and even fear of being replaced (Liberty and Zimmerman, 1986; Gilson, 1989). All these may motivate managers to manage earnings to fulfil their personal interests. The existing 'accrual accounting system'¹ gives managers a wide range of

¹ As per 'accrual accounting system', all the expenses and revenues pertaining to a financial year are recorded in the books of accounts irrespective of the fact, whether the expenses are actually paid or not and whether

discretion in determining their reported earnings. Managers do have adequate amount of control on the timing of actual expense items (Epps and Ismail, 2009) and they can even alter the timing of recognition of revenues and expenses (Toeh et. Al, 1998). If managers' incentives are dependent on financial performance of their firms, they may often try to manipulate earnings to fulfil their self- interests.

Why do financially distressed firms manipulate reported earnings is an important question to the investors, policy makers and regulators. Whatsoever, earnings management affects investors by reporting false information, due to which investors may not be able to make informed decisions. Toeh et al (1998), Rangan (1998) and dechow et al (1996) find evidence that managers inflate earnings prior to equity offerings. Earnings management can have significant impact on all stakeholders of a firm including the economy and society at large.

This raises the question as to what can protect the interest of the stakeholders with regard to earnings management. Here emerges the role of Corporate Governance system in place. Several studies in United Kingdom and United States show that, company boards influence earnings management and the quality of financial statements (Sarkar et al 2000). Cheng and Warfield (2005) provide the link between corporate governance and earnings management and find that, firms with higher managerial ownership do have lower level of earnings management. Studies also exhibit that independent boards reduce earnings management and are associated with better quality of financial reporting (e.g., Beasley [1996]; Dechow et al [1996]; Klein [2002]; Xie et al [2003]; Peasnell et al [2005]).

the revenues are actually received or not during that financial year. All Indian companies do follow 'accrual accounting system' for preparing their financial statements.

Against this backdrop, this paper makes an attempt to empirically examine the relationship between earnings management behaviour of financially distressed firms and their board characteristics to understand whether board quality affects earnings management. There are several motivations behind this study. First, there is an increasing trend in corporate defaults in the post financial crisis (2008) period in India. Second, there is acute dearth of studies examining the relationship between financial distress and earnings management in the Indian context. Third, to the best of our knowledge, this is the first study to examine whether board characteristics affect earnings management behaviour of financially distressed firms in an emerging market set up. The review of existing literature on the role of board characteristics on earnings management exhibit that, much of this literature are based on developed countries and very little work has been done on this issue on emerging economies, in spite of the fact that earnings management practices are more prevalent in emerging economies compared to the developed economies (Bhattacharya et al 2003).

Another interesting aspect is the ownership structure of Indian firms. Unlike widespread equity model in the developed economies, majority of Indian firms are owned or otherwise controlled by founding families (known as promoters). Inside owners of firms belonging to emerging economies may be motivated to manage earnings to conceal true financial pictures of firms from the outside world for extracting private benefits (Leuz, et al [2003]; Haw et al [2004]). In this relevance, policy makers, regulators, investors and other stakeholders have become sufficiently aware of the need for sound corporate governance system to ensure true and fair disclosure of corporate financial reports. This paper is expected to contribute to the literature by providing evidence in respect of earnings management behaviour of financially troubled firms and also the influence of board characteristics on earnings management. The

results are expected to be immensely beneficial to the investors, regulators, global managers and researchers with a special focus on an emerging economy.

The rest of the paper continues as follows. Section 2 deals with review of literature and hypotheses development. Section 3 explains research model, data and variables. Section 4 discusses the results of the study and finally section 5 provides concluding remarks.

2. Review of Literature and hypotheses development

2.1 Earnings management and financial distress

There have been a considerable number of studies exploring earnings management behaviour of the financially distressed firms around the world. Studies like Chen and Yuan (2004), Haw et al (2005), Chen et al (2006a, b), Yu et al (2006), Liu and Lu (2007) have examined whether financially distressed firms manipulate their earnings more frequently than normal firms in the Chinese market. DeFond and Jiambalvo (1994), Rosner (2003) and Beneish et al (2001) observe that, managers of firms which are under financial trouble, tend to shift earnings upward to avoid possible bankruptcy and debt covenant violations. Under such circumstances, managers usually focus on the short term survival of their firms (Charitou et al 2007b). Another set of studies argue that, managers of financially distressed firms are likely to shift their earnings downward in order to obtain better terms in their contract renegotiations (e.g., DeAngelo et al, 1994, Jaggi and Lee, 2002). These studies argue that, when firms are under severe level of financial distress, managers do reflect their true financial difficulties to renegotiate debt covenants with their creditors. Sometimes managers of distressed firms may report higher earnings at their self-interests like obtaining bonus, ensuring job security, enhancing compensation etc. (Agrawal and Chatterjee, 2015). While

examining the relationship between CEO incentives and earnings management, Bergstresser and Philippon (2006) observe that, more incentivised CEOs (those whose overall compensation is more sensitive to firms' share prices) lead to higher degree of earnings management. Studies have also analysed the earnings management behaviour of the firms in and around the global financial crisis in 2008. Strobl (2008) proposes a model indicating that earnings management is more prevalent during the times of economic booms and his proposition is supported by Cohen and Zarwin (2007). Chia et al (2007) observe that service sector firms based on Singapore were engaged in shifting their earnings downward during the crisis period.

2.2 Corporate Governance and earnings management

Corporate governance (CG) is primarily concerned with managing the business with accountability, fairness, transparency and responsibility so as to protect the interest of all stakeholders. Setting sound governance policy helps to avoid management level corruption and in creating shareholders' value. Agency theory explains the conflict of interests between managers and shareholders. Corporate governance mechanisms are expected to mitigate the agency cost by proper monitoring of the managers' activities and reducing opportunistic behaviour of the managers (Ashbaugh et al 2004). Board composition is considered to be the most important aspect of corporate governance. The challenge before the board is to ensure a balance between protecting shareholders' interests and simultaneously providing autonomy to the managers to run the firm effectively. This section of the paper concentrates on discussing the influences of boards' characteristics on earnings management and developing relevant hypotheses.

Board Independence

Existence of independent directors (ID_s) on board is considered to be an extremely important aspect of board composition. Outside independent directors are selected for protecting shareholders' interests (Rosenstein & Wyatt, 1990). As per corporate governance (CG) norms in India, at least 50 percent of the board should be independent for listed firms with executive chairperson, while for listed firms with non- executive chairperson, at least one third of the board should be independent. Since the ID_s do not have any direct or indirect benefits linked to the financial performance of firms, they are expected to facilitate reporting of true financial positions of the firms and thereby curb possibility of earnings manipulation, if any. Studies like Kiel & Nicholson (2003); Core et al (1999) do not find empirical evidence that independent outside directors create more effective boards and improve firm performance. However, Beasley (1996) finds that firms with higher proportion of ID_s are less likely to manipulate earnings. Peasnell et al (2005), with reference to UK firms, observe that ID_s do play effective role in curbing earnings management. Dechow et al (1996) provide evidence that firms with majority independent boards, smaller boards and an audit committee do prevent earnings management. Davidson et al (2005), on their study of 434 listed Australian firms, argue that majority of non-executive directors on the board and on the audit committee are found to be effective in reducing earnings manipulation. Hence, we hypothesize that, firms with higher proportion of ID_s are less likely to engage in earnings management.

H1: Firms with higher proportion of IDs on their boards have lower discretionary accruals.

Board Size

Effectiveness of the board often depends on its size. A smaller board is relatively easy to administer and becomes efficient. Prior research shows that, boards which are too large, lacks efficiency. Yermack (1996) finds evidence that smaller boards are more efficient and report better financial ratios. Huther (1997) finds inverse relationship between board size and firm performance in US. Rahman & Ali (2006) examine whether monitoring functions of board of directors, audit committee and concentrated ownership help in reducing earnings management, using 97 firms listed on the Main Board of Bursa Malaysia during 2002-03 and find that earnings management is positively associated with board size. In the Indian context, where promoters often occupy key managerial positions, it is likely that, earnings manipulation is relatively difficult when board size is large. If the board size is large, it may be relatively difficult to convince majority board members towards fulfilling the earnings manipulation motive, if any. Hence, we develop our second hypothesis as:

H2: Firms with large size of board of directors have less discretionary accruals.

Board Busyness and Board Diligence

Existing literature highlights potential benefits of multiple directorships. Studies find that, having directors on board with multiples appointments ensure better governance and constrain opportunistic earnings management. (e.g; Fama, 1980; Fama & Jensen, 1983). Another set of studies argue that too many multiple appointments may cause difficulties for independent directors to discharge their professional responsibilities effectively on boards (e.g; Ferris et al, 2003; Sarkat & Sarkar 2005). Over commitment by busy directors may reduce their abilities to attend board meetings, annual general meetings, and various committee meetings, which ultimately prevents independent directors to monitor the boards in the best

interest of the shareholders. Finding sufficient time to effectively discharge their professional responsibilities is a real challenge to the independent directors (Lipton & Lorsch, 1992).

Hence, we include another variable called '*board diligence*' as an additional important measure of board's quality. '*Board diligence*' implies the mean percentage of the board meetings attended by the independent directors over the year. Boards of the family owned corporations (widely prevalent in India) are often dominated by promoters with substantial ownership rights and occupy key managerial positions and control the firms (Carney & Gedajlovic, 2002; Gomes et al, 2003). Existing studies show that, these controlling shareholders do have incentives in engaging in opportunistic earnings management activities (Bertnard et al, 2002; Friedman et al, 2003; Fan & Wong, 2002). A diligent board is likely to monitor board's functions properly and thereby curb earnings manipulation. These discussions lead us to form our third and fourth hypotheses as:

H3: Firms with less 'busy boards' have lower discretionary accruals.

H4: Firms with higher 'board diligence' have lower discretionary accruals.

CEO duality

In corporate governance literature, the effectiveness of the board is viewed as a function of its independence, size and leadership structure (Epps & Ismail, 2008). Empirical studies on CEO duality have primarily focused on firms' performance evaluation and comparison and the results have been somewhat mixed. Braun & Sharma (2007) find no evidence behind the influence of CEO duality on firm performance in family controlled publicly traded firms. However, they observe that, when family ownership is less, CEO duality is beneficial measured by shareholders' returns. Brickley et al (1997); Chagnati et al (1993) and Rechner & Dalton (1991) provide evidence that, firms with separate chairpersons outperform firms with CEO

dominance. Jensen (1993) reports that the dual office structure also allows the CEO to control information available to the board and consequently impede effective monitoring. However, Xie et al (2003) find no relation between CEO dominance and earnings management. For Egyptian listed firms, Elsayed (2007) does not find any impact of CEO duality on firm performance. Dechow et al (1996) argue that firms whose CEOs are also the chairpersons of the boards, are more likely to be subject to accounting enforcement action by Securities and Exchange Commission (SEC) for any violation of the accounting principles. However, empirical evidence on the relationship between CEO duality and earnings management is very less, especially in emerging economies like India. Sarkar et al (2008) find that the presence of CEO-chairman affects the quality of the reported earnings of Indian firms. Based on these, we form our fifth hypothesis as:

H5: Firms where CEO is also the chairman of the board have higher discretionary accruals.

Promoters' influence

In emerging economies like India, family owned corporations dominate the industry landscape and the founding family members (promoters) often occupy key managerial positions with substantial ownership rights and control the firms (Carney & Gedajlovic, 2002). Hence it is important to examine whether the presence of controlling shareholders on firms' boards stimulates opportunistic earnings management. Existing literature highlights the incentives of controlling shareholders behind earnings management by camouflaging financial statements for their private benefits (Friedman et al, 2003, Bertnard et al 2002; fan & Wong, 2002). To measure the influence of promoters on board, we use a dummy variable which takes the value one if the controlling shareholders (promoters) are present on the board as executive directors and zero otherwise. Therefore our sixth hypothesis is:

H6: Firms with promoter influence on boards have higher discretionary accruals

3. Research design and measurement of variables

The sample for this study is comprised of financially distressed firms which have defaulted on their debt obligations and have been assigned a 'D' rating. The study period is from 2009 to 2014. The sample firms have been selected from four credit rating agencies namely, CRISIL Ltd., ICRA Ltd., CARE Ltd., and Fitch Ratings India Pvt. Ltd. The final sample comprise of 150 firms over the period 2009 to 2014. Data pertaining to various accounting variables, control variables etc. are procured from 'Prowess', a widely used and highly reliable corporate database, maintained by Centre for Monitoring Indian Economy (CMIE). Published annual reports and corporate governance reports of the sample firms available in 'Prowess' have been used for computing variables relating to board characteristics. The Prowess database has formed the basis of several published empirical studies on Indian corporate sector (e.g., Chatterjee & Mukherjee, 2015; Khanna & Palepu, 2000; Bertrand et al 2002).

3.1 Measurement of Earnings Management

Several studies have used discretionary accruals as proxy for earnings management (e.g.; Lu, 1999; Krishnan, 2003; Agrawal & Chatterjee, 2015; Siregar & Utama, 2008). Researchers use several models to measure discretionary accruals. Healy (1985), DeAngelo (1986), Jones (1991) and Dechow *et al.* (1996) are the most widely used models for measuring discretionary accruals. DeFond and Jiambalvo (1994) argue for the use of cross sectional data to estimate discretionary accruals in order to avoid the assumption of stability of coefficients and to reduce the probability of misspecification of the model. Guay *et al.* (1996) argue that Jones model and modified Jones model are the only models which provide evidence that is consistent with opportunism as well as performance measure hypothesis. However, Dechow *et al* (1995) observe modified Jones model outperforms Jones model. The modified Jones

model has been extensively used in literature for estimating discretionary accruals (Chen *et al.*, 2010; Aref *et al.*, 2012; Jagdish *et al.*, 2014; Toto *et al.*, 2014; Agrawal & Chatterjee, 2015; Jahmani, et al, 2016).

Based on the above, this paper uses discretionary accruals as proxy for earnings management by using the cross sectional modified Jones model. Separate industry-wise regressions have been run for estimating the parameters of the following model.

$$\frac{TA_{it}}{A_{it-1}} = \beta_1 \left(\frac{1}{A_{it-1}} \right) + \beta_2 \left(\frac{\Delta REV_{it} - \Delta REC_{it}}{A_{it-1}} \right) + \beta_3 \left(\frac{PPE_{it}}{A_{it-1}} \right) + \varepsilon_{it} \dots\dots\dots (Eq.1)$$

Where TA_{it} is total accruals of firm i in year t , estimated as the difference between net income of firm i in year t (NI_{it}) and cash flow from operations of firm i in year t (CFO_{it}). A_{it-1} denotes total assets of firm i in year $t-1$. ΔREV_{it} indicates change in revenues of firm i from year $t-1$ to year t and similarly ΔREC_{it} indicates change in receivables of firm i from year $t-1$ to year t . Finally PPE_{it} is Property, plant and equipment of firm i in year t . ε_{it} is the error term.

The estimated values of total accruals, which actually implies Non-Discretionary accruals (NDA_{it}), are obtained by putting the values of estimated parameters, β_1 , β_2 and β_3 , in the Equation 1 as follows:

$$NDA_{it} = \hat{\beta}_1 \left(\frac{1}{A_{it-1}} \right) + \hat{\beta}_2 \left(\frac{\Delta REV_{it} - \Delta REC_{it}}{A_{it-1}} \right) + \hat{\beta}_3 \left(\frac{PPE_{it}}{A_{it-1}} \right) + \varepsilon_{it} \dots\dots\dots (Eq.2)$$

Where NDA_{it} is non-discretionary accruals of firm i in time period t .

Finally, discretionary accruals are computed by subtracting non-discretionary accruals from the total accruals as follows:

$$DA_{it} = \frac{TA_{it}}{A_{it-1}} - NDA_{it} \text{ ----- (Eq. 3)}$$

Where DA_{it} is discretionary accruals for firm i in time period t

For analyzing the effect of board characteristics on earnings management, we have considered six board characteristics namely, (1) board independence, (2) board busyness, (3) board diligence, (4) board size (5) CEO duality (6) and Promoters' influence.

3.2 Control variables

Apart from board characteristics, various firm level factors may also impact earnings management behaviour of firms. Controlling of these factors is considered to be important in empirical researches to avoid any spurious association between earnings management and board characteristics. Existing studies find that size of a firm (*size*) is likely to affect earnings management (Charitou et al., 2011; Chen et al., 2010; Habib et al., 2013; Selahudin et al., 2014; Agrawal & Chatterjee, 2015). Studies like Barton & Simko, 2002; Habib et al., 2013 argue that larger firms are more likely to engage in earnings management compared to smaller firms due to more pressure to beat analysts' expectations and greater bargaining power with the creditors, while Agrawal & Chatterjee(2015) do not find any association between firm size and earnings management. The degree of financial leverage (*LEV*) is also likely to affect earnings management of firms (Charitou et al., 2011; Chen et al., 2010; Habib et al., 2013; Agrawal & Chatterjee, 2015). It is observed that highly levered firms tend to engage in greater earnings management (DeFond & Jiambalvo, 1994; DeFond & Park, 1997; Press & Weintrop, 1990). Earnings management is also likely to be influenced by Cash flow from operations (*CFC*) of firms (Charitou et al., 2011; Chen et al., 2010; Habib et al., 2013). Existing studies find a

negative relationship between cash flow from operations and earnings management (Becker et al., 1998; Kim et al 2003; Agrawal & Chatterjee, 2015). Firms with adequate and uninterrupted cash flows are less likely to engage in earnings management. Firms' profitability (*ROA*) and growth (*M/B Ratio*) also have been found to be associated with earnings management (Kothari et al, 2005; Skinner & Solan, 2002; Agrawal & Chatterjee, 2015). Nasuhiyah et al. (1994) observe that firms with low profit try to inflate earnings to report earnings above a certain level in order to avoid the risk arising out of earnings fluctuations. Chen et al, 2010 also finds similar results. Robin & Wu, 2012 argue that managers of high growth firms tend to manage earnings to signal future performance of their firms. Sarkar et al (2008) find that high growth firms are associated with higher earnings management. We consider these five variables as control variables in our regression.

Equity holdings by institutional investors is another important aspect to be considered specially with reference to emerging economies like India, where ownership is concentrated in the hands of promoters. In such type of ownership framework, institutional investors are likely to play a significant role in firms' governance and constrain earnings manipulation. Using a sample of U.S. firms over 1988-1996, Chung et al (2002) argue that large institutional shareholders constrain opportunistic earnings management. Mitra & Cready (2005) also find similar evidence. For examining the role of institutional investors in earnings management, if any, we include two more variables namely, equity shareholdings by domestic institutional investors (*DII_share*) and equity shareholdings by foreign institutional investors (*FII_share*). Domestic institutional investors comprise of Indian financial institutions, Indian banks and insurance companies and Indian mutual funds. All other institutional investors are categorized as foreign institutional investors.

Indian corporate sector is also characterised by the dominance of business groups. Several papers on ownership structure and earnings management exhibit that, where a large deviation in cash flow and control rights exists, controlling insiders do have incentives to conceal their non-maximising behaviour from the outside minority shareholders (Fand & Wong, 2002; Kim & Yi, 2005). To control this factor, we use a dummy variable, *group*, in order to capture the incentive of group affiliates to engage in earnings management. All regressions include industry dummies to capture unobserved industry effects, if any.

3.3 The Regression Model

The main cross sectional regression which we estimate, takes the following form:

$$\begin{aligned}
 DA = \alpha &+ \beta_1 \times \text{board independence} + \beta_2 \times \text{board busyness} + \beta_3 \times \text{board diligence} \\
 &+ \beta_4 \times \text{board size} + \beta_5 \times \text{CEO duality} + \beta_6 \times \text{promoters' influence} \\
 &+ \beta_7 \times \text{DII share} + \beta_8 \times \text{FII share} + \beta_9 \times \text{Group} + \beta_{10} \times \text{Size} + \beta_{11} \\
 &\times \text{Lev} + \beta_{12} \times \text{CFC} + \beta_{13} \times \text{ROA} + \beta_{14} \times \text{MB Ratio} \\
 &+ \text{Industry dummies} + \text{error}
 \end{aligned}$$

Table 1 summarizes the list of variables considered in our models along with their definitions.

Table 1: List of variables and definitions

Variable Name	Definition
Maj_indep_board	Majority Independent board. Dummy variable. Equals one if the percentage of independent director exceeds 50, and zero otherwise.
Percent_ID	Percentage of independent directors on the board. Computed as number of independent directors divided by total board size, multiplied by 100.
Board_busyness	Board Busyness. Dummy variable. Equals one if the board has at least one independent director with three or more directorships, and zero otherwise.
Board_diligence	Board diligence. Calculated as mean of percentage of board meetings attended by independent directors.
Board_size	Size of the board. Calculated as total number of directors on the board at the end of the financial year.
CEO_duality	CEO duality. Dummy variable. Equals one if the CEO is also the chairman of the board, and zero otherwise.
Promoter_influence	Promoter influence. Dummy variable. Equals one if controlling shareholders or promoters are present on the boards in the capacity of executive director, and zero otherwise.
DII_share	Domestic institutional investors' share. Computed as percentage of equity owned by Indian financial institutions, Indian mutual funds, Indian banks and insurance companies.
FII_share	Foreign institutional investors' share. Calculated as percentage of equity owned by institutional investors that are not domestic institutional investors.
Group	Dummy variable. Equals one if the company is affiliated to a business group, and zero otherwise.
Size	Measured by natural log value of total assets
LEV	Degree of financial leverage derived as long term debt divided by total assets
MB Ratio	Market to book value ratio, proxy for growth of the firm. Calculated as ratio of market value of equity over book value of assets.
CFC	Cash flow coverage. Derived as cash flow from operations plus cash and cash equivalents, divided by interest expense.
ROA	Measurement of firms' profitability. Computed as net income divided by total assets, multiplied by 100.

4. Empirical Findings

4.1 Summary Statistics

The summary statistics for our sample are presented in Table 2. The mean value of discretionary accruals (DA) is positive, about 0.20 percent of the lagged total assets, which implies that, firms on an average, are engaged in income increasing earning management. However, the maximum value (0.279) and minimum value of DA (-0.472) indicate that firms are engaged in both income increasing as well as income decreasing earnings management. Firm size has a mean value of 8.69, while mean leverage is 0.47. The median values of these two variables are almost equal to their respective mean values, indicating that sample provides symmetric representation to firms of different sizes and leverage. A wide range of difference in CFC is evident from the maximum value (46.71) and minimum value (-4.52) of CFC with a high standard deviation (4.58). This implies that CFC varies widely across sample companies. Similar scenario is evident for firms' profitability (measured through ROA) as well. The negative mean value of ROA (-10.5%) signifies that sample firms, on an average, are loss making. While 39 percent of the sample firms are group affiliated, the mean value of the market to book value ratio is 0.31.

With reference to board composition, Table 2 shows that about 43 percent of our sample firms have majority independent boards, with mean percentage of independent directors being about 53 percent. 68 percent of the firms have busy independent directors on their boards, while the independent directors have attended about 75 percent of the board meetings. While the average board size is 7, there are firms with small boards (3 members) and firms with relatively large boards (14 members) in the sample. About 40 percent of the firms have promoters as inside directors on their boards. Nearly 56 percent of firms report

CEO duality. Referring to equity ownership, we find that domestic institutional investors own only about 3 percent of equity of the sample firms, while foreign institutional investors hold about 4 percent of total equity. For both DII share as well as FII share, the mean values are higher than the median values, implying that both types of institutions have large equity holdings in a few firms. DII share in some firm is as high as 25 percent and FII share is as high as 36 percent.

4.2 Regression Results

In our models, we estimate our regression of discretionary accruals on board composition under two different specifications. Under the first specification, the effect of board composition on discretionary accruals is examined by incorporating all control variables excluding those which are related to institutional investors. Under the second specification, institutional investors, both domestic as well as foreign, are introduced in regression models, given their relevance in emerging economy settings. This will contribute in two ways. First, by doing so, we can examine whether presence of institutional investors influence the relationship between board composition and earnings management. Second, it will help in examining whether institutional investors play any role in curbing earnings manipulation.

4.2.1 Board Composition and earnings Management

Table 3 presents the regression results for models 1-4. Here we estimate four different regression models to examine the effect of board composition on discretionary accruals without institutional ownership. In Models 1 and 2, we examine the impact of independence, busyness, diligence and size. In Models 3 and 4, we examine the influence of independence, busyness, diligence, size and the influence of promoters and the CEO.

The results of Model 1 and 2 show that board independence under both the measures i.e.; majority independent board and percentage of independent directors on boards, have no impact on discretionary accruals (p values of 0.602 and 0.406 respectively). However, busyness of independent directors reduces earnings management, the coefficients are negative and are statistically significant at 5 percent level in both the models. Also we find that diligence of independent directors reduce discretionary accruals as the coefficients are negative in both the models and are statistically significant at 1 percent level. Board size is not found to have any significant impact on discretionary accruals on either model (p values of 0.46 and 0.46 respectively). From the results of model 3 and 4, we find (in addition to earlier results) that, promoters' presence on the board positively influence discretionary accruals, the coefficients are statistically significant at 5 percent level in both the models 3 and 4. Also CEO duality (where CEO of the firm is also the chairman of the board), is found to be positively associated with discretionary accruals under both the models, indicating that CEO duality leads to increase in earnings management. This finding is intuitive enough because dual chairman-CEOs often enjoy considerable power on the boards and hence may take decisions linked to self-interests rather than looking at the interest of the outside investors.

In a nutshell, from our regression results (Models 1-4) we find that, it is the diligence of the independent directors that reduces earnings management and not the number or the proportion of independent directors on the board. We also find that busy boards are effective in curbing earnings management. This finding is opposite to what we hypothesised. The possible explanation behind this finding could be, busy independent directors are competent enough to contribute on the governance and operations of firms effectively and this is why they are busy on their professional endeavours. While the presence of CEO duality results in increasing discretionary accruals, promoters' presence on boards also increase earnings management. Board size does not influence the earnings management behaviour of the firms.

4.2.2 Role of Institutional Investors

We extend our regression analysis of board characteristics on discretionary accruals after incorporating domestic and foreign institutional shareholdings and results are presented in Table 4. In Model 5, we include domestic institutional shareholdings, in Model 6, we consider foreign institutional shareholdings and in Model 7 we include both domestic as well as foreign institutional shareholdings. Finally in Model 8, we consider the joint effect of both types of institutional investors and promoter's influence on board along with all board characteristics and other control variables. Thus, Model 8 is our full model examining the relationship between board composition and earnings management.

From the regression results of Model 5 and 6, we find that various board characteristics continue to show similar results as earlier (as shown in Models 1-4)². Board independence does not affect discretionary accruals, while busy boards and diligent independent directors reduce discretionary accruals. While the board size does not impact discretionary accruals, CEO duality leads to increase in earnings manipulation and promoters' presence on boards as inside directors increase earnings management. The coefficients of domestic institutional ownership in Model 5 and foreign institutional ownership in Model 6 are not found to be statistically significant (p values of 0.34 and 0.28 respectively), signifying that neither domestic nor foreign institutional investors influences discretionary accruals and they are not associated with reduced earnings management. The results remain unchanged in Model 7 as well which considers both types of institutional shareholdings. This finding is contrary to the findings of Chen et al (2002) and Mitra & Cready (2005) who exhibit that large institutional shareholders constrain opportunistic earnings management.

Above results show that institutional shareholders by themselves do not have any significant impact on discretionary accruals. However, it is likely that powerful institutional investors can act as catalyst in mitigating the influence of controlling promoters on boards' effectiveness. Large and powerful institutional shareholders are effective in mitigating the agency costs between controlling shareholders and outside shareholders especially in the context of family owned corporations (Anderson & Reeb, 2003, Sarkat et al 2008). Applying this reasoning in an emerging market context (India), where family owned corporations

² Here we have used percentage of independent directors as a proxy for board independence to report the results in Table 4. Using majority independent boards also, all the results remain unchanged.

dominate the corporate landscape, one can argue that presence of institutional investors can constrain opportunistic earnings management.

To address this issue, following Sarkar et al (2008), we analyse the joint effect of institutional shareholders and promoters' presence on discretionary accruals. For estimating the joint effect, we incorporate two interaction terms in our model – interaction of domestic institutional ownership and promoter's presence as well as interaction of foreign institutional ownership and promoter's presence. Model 8 in Table 6 reports the results. Model 8 shows that the coefficient of the interaction term linked to domestic institutional investors is negative and statistically significant at 5 percent level (p value 0.043), signifying that domestic institutional investors curb opportunistic earnings management subject to promoters' influence. In this context, it is pertinent to mention that while promoter's influence on boards increases discretionary accruals (evident in Model 5-7), domestic institutional ownership mitigate the adverse impact of promoter's influence on earnings management of firms. However, the coefficient of the interaction term associated with foreign institutional ownership is found to be statistically insignificant (p value 0.296) suggesting that, foreign institutional investors do not play any compensatory role unlike domestic institutional investors. The difference between the results can be explained by the fact that, domestic institutional investors especially banks, financial institutions etc. can keep their nominees on the firms' boards to participate in the operations of firms. However, for an outsider, it may be difficult to assess the effectiveness of the governance mechanisms in place. Very few studies in corporate governance literature have examined the joint effect of two alternative mechanisms (e.g; Cremers & Nair, 2003; Sarkar et al, 2008). Our finding that there is a mitigating effect on earnings management in the presence of both, has

important implications for regulators while framing their policies as well for investors while taking investment decisions

4.2.3 The Impact of other control variables

Model 7 of Table 6 reveals that size is negatively associated with discretionary accruals, the coefficient is negative and is statistically significant at 5% level. This indicates that larger firms are less prone to earnings manipulation. Market to book value ratio, a proxy for firms' growth, is positively associated with discretionary accruals (p value 0.037), indicating that high growth firms are associated with higher earnings manipulation. The coefficient of CFC is found to be negative and is statistically significant at 1% level (p value of 0.000). This simply indicates that, firms with higher cash flow coverage have less incentives to manipulate their earnings. Becker et al (1998), Kim et al (2003) also find similar results. Firms' profitability, measured through ROA, is found to be positively related to discretionary accruals (p value of 0.000), indicating that profitable firms tend to manage their earnings. Thus is so because, managers of less profitable firms wish to achieve their target earnings level to fulfil their self-interests and / or to ensure that investors do not lose confidence in the firm, whereas, firms with severe distress level do not inflate their earnings, rather they prefer to disclose their true financial difficulties in order to obtain better terms from their creditors and lenders. Nasuhyah et al. (1994), Jaggi and Lee (2002) and Chen et al. (2010) also find similar results. The coefficients of leverage and group affiliation are statistically insignificant (p values of 0.27 and 0.29 respectively), implying that they do not have any impact on discretionary accruals. Finally we find the existence of industry effect on discretionary accruals.

5. Conclusion

This paper examines whether board characteristics affect earnings management behaviour of the financially troubled firms in the context of a large emerging economy, India. By considering a wide range of board characteristics namely, independence, busyness, diligence, size, promoters' influence and CEO duality, this study contributes to the literature as to whether the governance mechanisms that are effective in developed economies are equally important in the Indian context too. Using a sample of 150 financially distressed firms over the period 2009 to 2014 and applying discretionary accruals as a proxy for earnings management, the study finds that it is the board quality that actually helps in curbing earnings manipulation and not just the board independence. Our results indicate that board independence has no significant impact on earnings management rather diligent boards help in reducing earnings management. As opposed to the general perception, busy independent directors are found to be effective in curbing earnings management. While the presence of CEO duality affects the quality of reported earnings, promoters' influence on boards increase earnings management. However, Board size does not influence earnings management behaviour of the firms. We also find that, in the existence of various board characteristics, domestic or foreign institutional investors do not have any independent effect on earnings management. However, interaction between promoter's presence and domestic institutional ownership reduces earnings management, which means the adverse impact of promoter's presence on boards is effectively mitigated by the presence of domestic institutional investors. Finally, the study establishes the fact that, a carefully designed board with certain qualities is helpful in curbing opportunistic earnings management behaviour of firms in the emerging market context too.

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Table 2
Descriptive Statistics (Full Sample)

Variable	Mean	Median	Std. Deviation	Min	Max
DA	0.002	0.001	0.093	-0.472	0.279
Maj_indep_board	0.427	0.000	0.496	0.000	1.000
Percent_ID	52.549	50.000	11.449	11.110	80.000
Board_busyness	0.680	1.000	0.468	0.000	1.000
Board_diligence	75.497	78.350	17.689	8.330	100.000
Board_size	7.467	7.000	2.342	3.000	14.000
Promoter_influence	0.4129	0.000	0.3721	0.000	1.0000
CEO_duality	0.564	1.000	0.498	0.000	1.000
Group	0.387	0.000	0.489	0.000	1.000
Size	8.695	8.670	1.371	5.727	12.052
Lev	0.469	0.483	0.175	0.080	1.151
MB_ratio	0.313	0.159	0.509	0.000	3.061
ROA	-0.105	0.979	7.436	-32.636	47.276
CFC	1.580	0.856	4.585	-4.522	46.714
DII_share	2.591	0.415	4.159	0.000	24.900
FII_share	3.585	0.080	6.478	0.000	36.030

Table 3
Effect of Board Composition on Discretionary Accruals

Explanatory Variables	Expected Sign	Model 1	Model 2	Model 3	Model 4
Intercept		0.0511*** (0.000)	0.0800*** (0.000)	0.0692*** (0.000)	0.0871*** (0.000)
Maj_indep_board	-	-0.0080 (0.602)		-0.0070 (0.503)	
Percent_ID	-		-0.0010 (0.406)		-0.0010 (0.545)
Board_busyness	+	-0.0020** (0.041)	-0.0030** (0.042)	-0.0031** (0.049)	-0.0030** (0.043)
Board_diligence	-	- 0.0009*** (0.0046)	-0.0009*** (0.0039)	-0.0009*** (0.0041)	-0.0009*** (0.0041)
Board_size	-	0.0015 (0.458)	0.0012 (0.461)	0.0012 (0.421)	0.0016 (0.394)
Promoter_influence	+			0.0033** (0.0493)	0.0034** (0.039)
CEO_duality	+			0.015** (0.021)	0.014** (0.019)
Group (ownership)	+	-0.0021 (0.381)	-0.0030 (0.333)	-0.0023 (0.343)	-0.0041 (0.314)
Size	-	-0.0247 (0.500)	-0.0240 (0.487)	-0.0231 (0.514)	-0.0261 (0.503)
Lev	+	0.0041 (0.397)	0.0049 (0.311)	0.0044 (0.430)	0.0047 (0.433)
MB Ratio	+	0.0070** (0.044)	0.0070** (0.041)	0.0082** (0.040)	0.0080** (0.048)
ROA	-	0.0060*** (0.000)	0.0061*** (0.000)	0.0060*** (0.000)	0.0062*** (0.000)
CFC	-	-0.0080*** (0.000)	-0.0081*** (0.000)	-0.0080*** (0.000)	-0.0083*** (0.000)
Industry dummies		YES	YES	YES	YES
No of Observations		150	150	150	150
R ²		0.2715	0.2731	0.2747	0.2757
F		4.247	4.295	4.612	4.628
Prob. F		0.000	0.000	0.000	0.000

Table 4
Effect of Board Composition and Institutional Investors on Discretionary Accruals

Explanatory Variables	Expected Sign	Model 5	Model 6	Model 7	Model 8
Intercept		0.0870*** (0.000)	0.0601*** (0.000)	0.0690*** (0.000)	0.0602*** (0.000)
Percent_ID	-	0.0001 (0.5460)	0.0002 (0.6201)	0.0001 (0.6105)	0.0002 (0.6223)
Board_busyness	+	-0.0040** (0.0341)	-0.0032** (0.0293)	-0.0031** (0.0211)	-0.0050** (0.0317)
Board_diligence	-	- 0.0009*** (0.0032)	- 0.0008*** (0.0029)	-0.0008*** (0.0025)	- 0.0008*** (0.0030)
Board_size	-	-0.0216 (0.3952)	-0.0235 (0.4124)	-0.0242 (0.3821)	-0.0231 (0.4032)
Promoter_influence	+	0.0048** (0.0325)	0.0046** (0.0441)	0.0042** (0.0437)	0.0029** (0.0154)
CEO_duality	+	0.0140** (0.0251)	0.0170** (0.0279)	0.0161** (0.0216)	0.0270** (0.0426)
DII_share	-	0.0011 (0.3411)		0.0012 (0.3621)	0.0011 (0.3052)
FII_share	-		-0.0019 (0.2752)	-0.0018 (0.2934)	-0.0019 (0.2723)
Promoter_influence × DII_share	?				-0.0016** (0.0425)
Promoter_influence × FII_share	?				-0.0004 (0.2961)
Group (ownership)	+	-0.0019 (0.3612)	-0.0021 (0.3184)	-0.0020 (0.3561)	-0.0035 (0.2922)
Size	-	-0.0031 (0.1954)	-0.0052** (0.0381)	0.0051** (0.0324)	-0.0061** (0.0432)
Lev	+	-0.0035 (0.4211)	-0.0039 (0.3862)	-0.0038 (0.3590)	-0.0057 (0.2710)
MB Ratio	+	0.0081** (0.0300)	0.0070** (0.0362)	0.0072** (0.0292)	0.0080** (0.0375)
ROA	-	0.0061*** (0.0000)	0.0057*** (0.0000)	0.0051** (0.0000)	0.0069*** (0.0000)
CFC	-	-0.0080*** (0.0000)	-0.0080*** (0.0000)	0.0071*** (0.0000)	-0.0081*** (0.0000)
Industry dummies		YES	YES	YES	YES
No of Observations		150	150	150	150
R ²		0.2761	0.2822	0.2751	0.2844
F		4.381	4.463	4.427	4.245
Prob. F		0.000	0.000	0.000	0.000

